Kuldip Karangiya

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AI/ML Developer
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AI/ML and Backend Developer with approximately 2 years of hands-on experience designing and deploying production-grade machine learning systems and scalable backend services. Experienced in building HIPAA-compliant healthcare platforms, asynchronous data pipelines, and LLM-powered data extraction workflows using Python and FastAPI. Skilled at translating complex medical and analytical data into reliable, high-performance solutions that deliver measurable business and clinical impact.

Skills

Programming & Backend	Python, FastAPI, SQLAlchemy, Pydantic, REST APIs, WebSockets, Async Programming
AI / Machine Learning	Machine Learning (classification, recommendation, forecasting), LLMs (GPT-4), RAG pipelines, Multi-Agent Systems, Prompt Engineering.
NLP & Computer Vision	NLP: LangChain, LangGraph, SpaCy, NLTK Computer Vision: YOLO, ResNet50.
Databases & Cloud	PostgreSQL, Redis, Alembic, Google Cloud Platform (Cloud Run, GCS), Docker, GitHub Actions.
Testing & Tooling	Pytest (Unit, Integration, E2E), Mocking, Swagger / OpenAPI, Playwright, pyecharts, Streamlit.

Experience

Data Scientist | Bitontree Solutions, Ahmedabad **Jul 2025 – Present**

- Led the development of a HIPAA-compliant healthcare API platform supporting 15,000+ patients and processing 10,000+ medical records per month with 99.9% uptime.
- Owned end-to-end backend delivery for AI-driven healthcare workflows, ensuring reliable orchestration across ingestion, analysis, and report generation services.
- Designed and maintained scalable asynchronous processing pipelines to handle high-concurrency workloads with consistent sub-200 ms API latency.
- Implemented clinical data quality and deduplication workflows, significantly improving biomarker consistency across longitudinal patient records.
- Delivered personalized medical visualizations and reports to end users, improving performance and user experience through optimized backend services.
- Established production-grade CI/CD pipelines using Docker and GitHub Actions, reducing deployment time and improving release reliability across environments.

AI/ML Developer | AUM Imagineering Pvt. Ltd., Ahmedabad | Dec 2023 – July 2025

- Deployed three production-grade ML models (recommendation, classification, forecasting) on Azure, achieving 99.5% uptime and reducing manual inference time by 60%.
- Optimized NLP pipelines using SpaCy and custom scripts to process 50,000+ queries daily, cutting preprocessing latency from 800 ms to 250 ms.
- Designed and implemented a LangChain-powered chatbot handling 10+ concurrent farmer queries, increasing weekly active users by 40%.
- Integrated real-time weather and subsidy data feeds via REST APIs, improving advisory accuracy by 30%.
- Mentored two interns on model evaluation and tuning, resulting in a 20% lift in F1-score across key use cases.

Projects

N1 Healthcare | Jun 2025 – Present

- Architected a FastAPI-based orchestration layer for AI-driven healthcare workflows using an event-driven, asynchronous design.
- Built a high-throughput background processing pipeline using Redis Streams to manage medical record ingestion, preprocessing, and report generation under concurrent workloads.
- Designed a biomarker canonicalization and analytics system using SHA-256 hashing and PostgreSQL advisory locks to ensure data consistency across longitudinal patient records.
- Implemented secure file ingestion and storage workflows using Google Cloud Storage with signed URLs and background processing for large medical documents.
- Developed a dynamic visualization engine using pyecharts and Playwright to generate personalized biomarker charts (SVG/PNG/PDF), leveraging caching to optimize rendering performance.
- Implemented automated Markdown-to-PDF report generation with optional PDF-to-DOCX conversion to support configurable and multilingual clinical reports.

Intelligent Farm Advisory Chatbot | Dec 2024 – Mar 2025

- Built and deployed an NLP/LLM chatbot that handled 5,000+ farmer queries per month, achieving 85% intent-recognition accuracy and reducing manual support tickets by 70%.
- Integrated real-time weather and subsidy feeds, cutting average farmer wait-time from 2 hours to under 5 minutes.

Tiles Recommendation System | Jan 2024 – May 2024
- Implemented a ResNet50 + k-NN pipeline, increasing recommendation precision from 72% to 89% and boosting user click-through by 35%.
- Processed a catalog of 20,000+ images, reducing query response time from 1.2 s to 350 ms.
- Developed a Streamlit interface enabling seamless user interaction and visual comparison of recommended tiles.

LinkedIn Parser | Jan 2023 – Dec 2023
- Parsed 50,000+ LinkedIn profiles, automating candidate shortlisting and cutting manual screening time by 60%.
- Trained and deployed a ranking model that improved recruiter match-rate from 40% to 68%, accelerating hiring cycles by 25 days on average.

Education

B.Tech in Agricultural Information Technology
Anand Agricultural University, Anand 2020–2024

High School in Science
Delta Science School 2018–2020

Activities/Certifications

- Coursera: Crash Course on Python.
- University-level 4×100m relay gold medalist (2022–23).
- University-level 100m silver medalist (2022–23).
- University-level 200m silver medalist (2022–23).